Statistical Machine Learning - HW 5

Problems 1-4: Chapter 6

- **Exercise #1, #3, #4, #10** from Chapter **6**

Problems 5-6: Chapter 7

- **Exercise #3, #6** from Chapter **7**

Problem 7: Chapter 9 (SVM)

E.1 Soft-margin SVM

In real world applications, there are outliers in data. This can be dealt with using a soft margin, specified in a slightly different optimization problem as below (soft-margin SVM):

$$\min \frac{1}{2}\mathbf{w}\cdot\mathbf{w} \;+\; C\sum_{i=1}^{N}\xi_i \quad \text{where } \xi_i \geq 0$$

subject to

$$y^{(i)}(\mathbf{w}^T\mathbf{x}^{(i)} + b) \geq 1 - \xi_i.$$

ξ_i represents the slack for each data point i, which allows misclassification of datapoints in the event that the data is not linearly seperable. SVM without the addition of slack terms is known as hard-margin SVM.

1. Intuitively, where does a data point lie relative to where the margin is when $\xi_i = 0$? Is this data point classified correctly?
2. Intuitively, where does a data point lie relative to where the margin is when $0 < \xi_i \leq 1$? Is this data point classified correctly?
3. Intuitively, where does a data point lie relative to where the margin is when $\xi_i > 1$? Is this data point classified correctly?

E.2 Kernel SVM

Support Vector Machines can be used to perform non-linear classification with a kernel trick. Recall the hard-margin SVM from class:

$$\min \frac{1}{2}\mathbf{w}\cdot\mathbf{w} \quad \text{s.t.} \quad y^{(i)}(\mathbf{w}^T\mathbf{x}^{(i)} + b) \geq 1.$$

The dual of this primal problem can be specified as a procedure to learn the following linear classifier:

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i y_i (\mathbf{x}_i^T \mathbf{x}) + b.$$

Note that now we can replace $\mathbf{x}_i^T\mathbf{x}$ with a kernel $k(\mathbf{x}_i, \mathbf{x})$, and have a non-linear decision boundary.

In Figure 1, there are different SVMs with different shapes/patterns of decision boundaries. The training data is labeled as $y_i \in \{-1, 1\}$, represented as the shape of circles and squares respectively. Support vectors are drawn in solid circles.

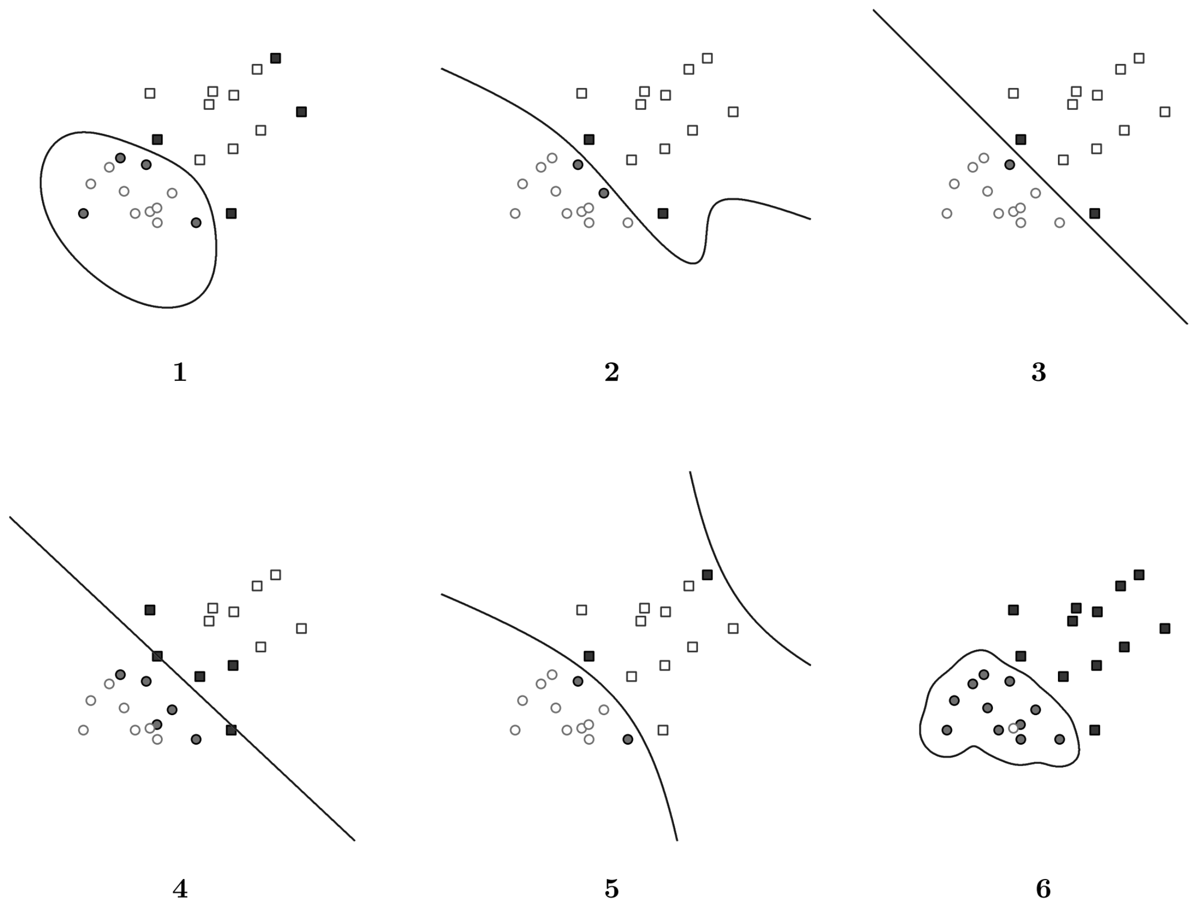

Figure 1: Figure 1: SVM boundaries

Match the scenarios described below to one of the 6 plots (note that one of the plots does not match to anything). Each scenario should be matched to a unique plot. Explain in less than two sentences why it is the case for each scenario.

1. A hard-margin linear
2. A soft-margin linear
3. A hard-margin kernel SVM with $k(u, v) = u \cdot v + (u \cdot v)^2$
4. A hard-margin kernel SVM with $k(u, v) = \exp\left(-5\|u - v\|^2\right)$
5. A hard-margin kernel SVM with $k(u, v) = \exp\left(-\frac{1}{5}\|u - v\|^2\right)$